



[Best trails in Van Anda](#)

Rainwater Harvester (RWH) System Design

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Introduction

- Communities lack access to safe drinking water
- A Rainwater Harvesting (RWH) System provides reliable water access



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Project
Overview

Problem Context

Objectives:

- To design a Rainwater Harvester (RWH) system for a remote, off-grid household
- Maximize overall system satisfaction

Main Design Criteria:

- System reliability and cost
- Maintenance requirement
- Sustainability
- Flow rate

Stakeholder Benefits

- Provide a consistent and reliable water supply to Van Anda
- Optimize cost over system lifetime
- Lower environmental impact
- Manage maintenance requirements



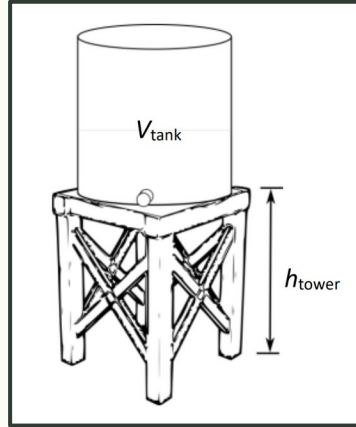
Final Design Recommendation



Pump 3



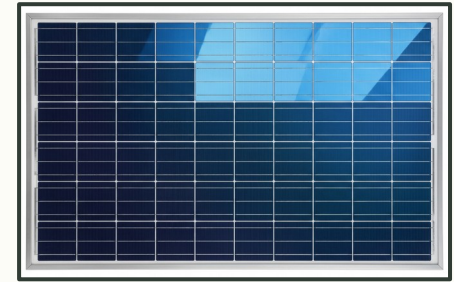
**All filters present
before storage tank**



**Size: 25 m^3
Location: (40, -10,
18)**



**Chlorine Disinfection
(and 50W UV)**



Solar (HES-305P)

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RWH System Design Process

Overall Strategy

1. Start with a selection of components that produce moderate satisfaction
2. Switch components and observe the change in satisfaction until the maximum



Key Considerations

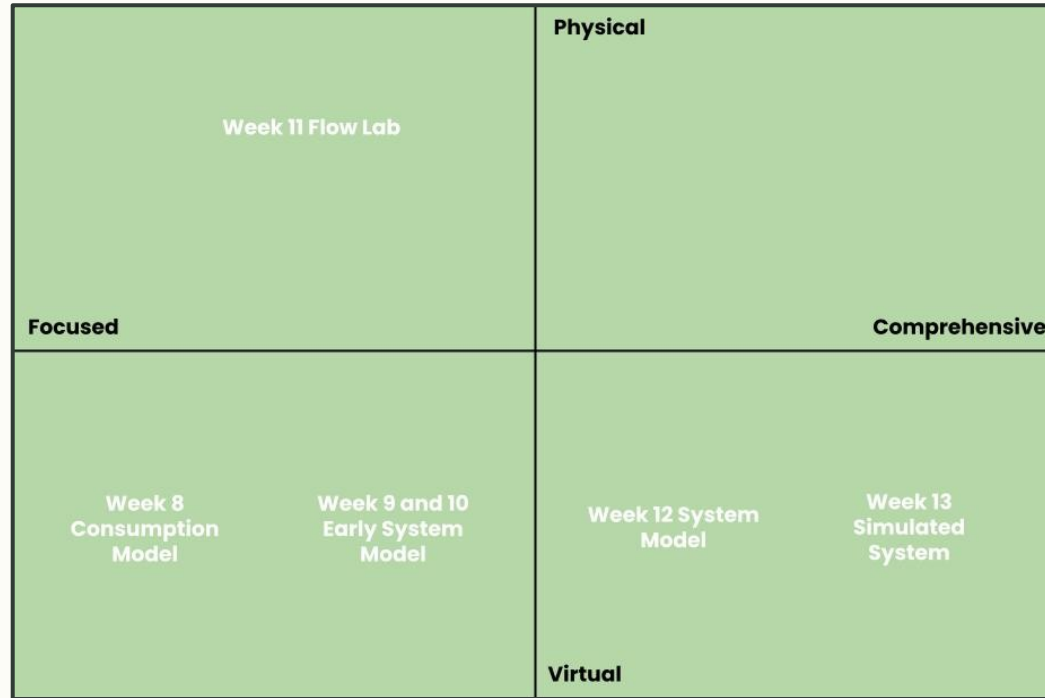
- Costs
- Consumption
- Health and Environment Exposure
- Power System and Greenhouse
- Maintenance Occurrence
- Non-Potable Water Supply
- On-Demand Flow Rate
- Reliability



Key Assumptions

- Same amount of water used each day
- Water standard will be met
- Water density remains constant
- Contact time of ozone and chlorine is 10 minutes
- Daylight hours

Types of Prototypes

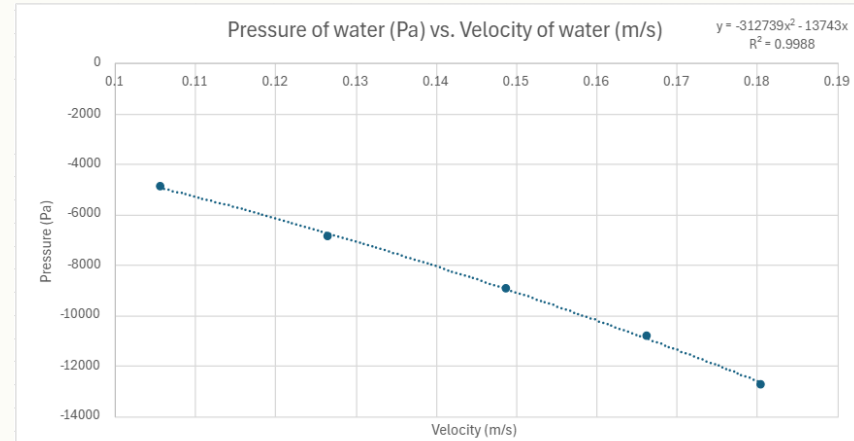


Pump Prototype and Filter Coefficient

Observations:

- With greater height, the flow rate increased
- Increasing length of outlet decreases flow velocity
- Above a certain height, the pump fails to pump water out

Cf was determined through curve fitting these data, at 13743



Final Component Selections and Key Tradeoffs

1. Pump Choice

Choice: Pump 3

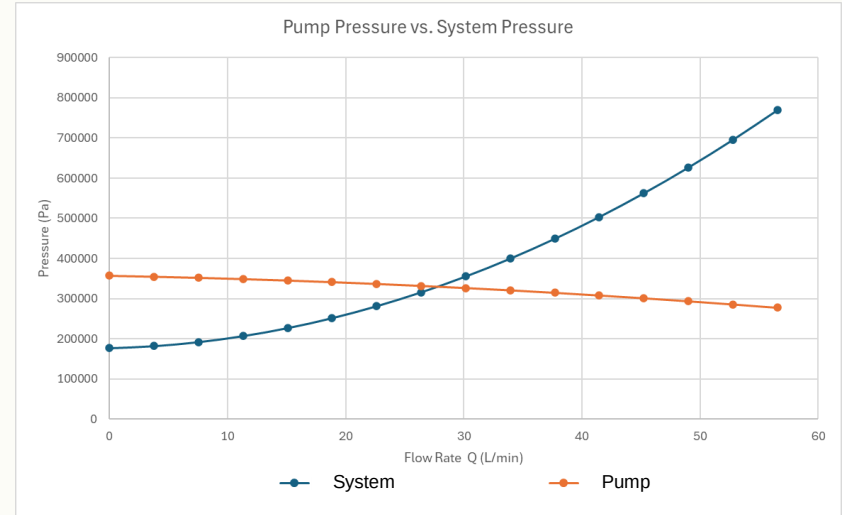
- Higher efficiency

Considerations:

- Pump 2 and 3 produce the same flow rate

Trade off:

- Cost



2. Storage Tank Size and Location

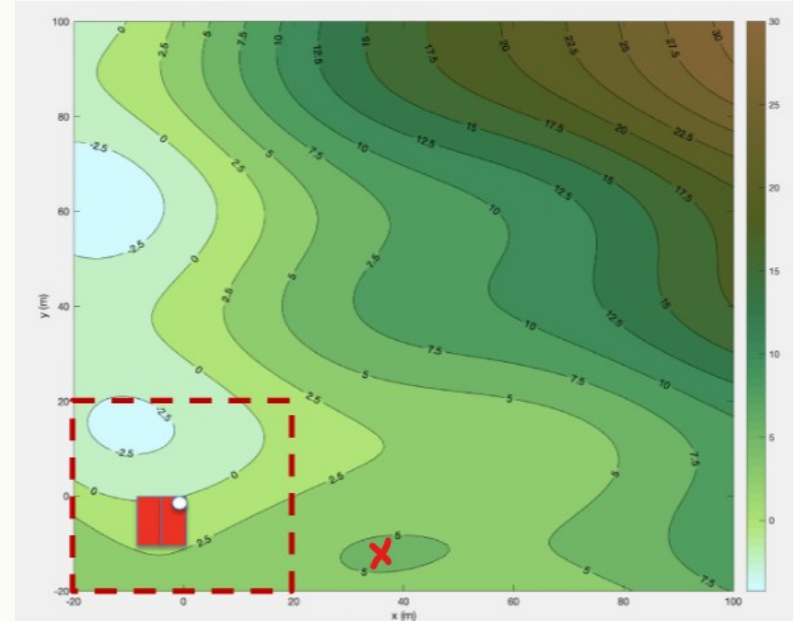
Choice:

- Size: 25 cubic metres
- Additional Storage Tower Height = 13 m

Reason:

- Ideally tank closeby with additional height

Trade off: Variable efficiency at 38%



3. Filter Placement

Choice: All three filters present (1 μm , 5 μm , 100 μm) before storage tank

Reason: Increases the lifespan of each, reduces total price and maintenance

Trade off: Higher initial costs

4. Power System

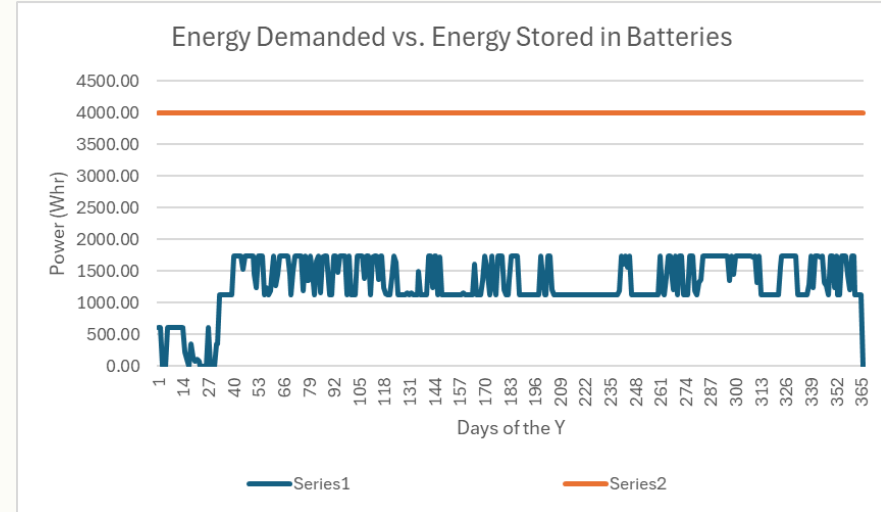
Choice: Solar Panel HES-305P (3 panels)

Reason:

- Good Efficiency
- Low health and environmental Risk

Trade off:

- Limited by sun hours and affected by weather



5. Disinfection System

Choice: Chlorine and 50W UV System

Reason: - Significantly cheaper than ozone in terms of energy
- A 50W UV system is cost effective for our flow rate of 30 LPM

Trade off: - Chlorine introduces health risk



6. Non-Potable Water Supply

Choice: 0.3% of Consumption

Reason:

- Many days water stored would fill capacity
- Minutely affected reliability



Alternative Configurations for Consideration

Comparison:

- Diesel and Ozone has the highest cost of \$74,481, and only receive 1% satisfaction in Greenhouse Emissions
- Greater number of solar panels required to meet ozone requirements
- Diesel and Chlorine has the second highest cost of \$62,157
- Solar and Chlorine has the cheapest cost, while meeting all other requirement

Stakeholder Satisfaction

MULTI-CRITERIA PERFORMANCE SATISFACTION MATRIX								
	Attribute	Parameter	Equation	Value	Min Value	Max. Value	Satisfaction	Weighted Percents
	Consumption	C	$S(C) = \frac{1}{2} \left(1 - \cos \left(\frac{C - C_{\min}}{C_{\max} - C_{\min}} \pi \right) \right)$	600.00	125	745	0.8710068	0.139361087
	Relative Cost	C_R	$S(C_R) = \frac{1}{2} \left(1 + \cos \left(\frac{C_R - C_{\min}}{C_{\max} - C_{\min}} \pi \right) \right)$	59,347.33	33000	126500	0.816541	0.155142785
	Health & Environmental Risk resposure	E	$S(E) = \frac{1}{2} \left(1 + \cos \left(\frac{E - E_{\min}}{E_{\max} - E_{\min}} \pi \right) \right)$	-	1	24	1	0.1175
	Power System & GHG	G	$S(G) = \frac{1}{2} \left(1 + \cos \left(\frac{G - G_{\min}}{G_{\max} - G_{\min}} \pi \right) \right)$	49.85	15	110	0.7031898	0.091414671
	Maintenance Occurence	M	$S(M) = \frac{1}{2} \left(1 + \cos \left(\frac{M - M_{\min}}{M_{\max} - M_{\min}} \pi \right) \right)$	15.00	10	60	0.9755223	0.13573956
	Non-Potable Water Supply	NP	$S(NP) = \frac{1}{2} \left(1 - \cos \left(\frac{NP - NP_{\min}}{NP_{\max} - NP_{\min}} \pi \right) \right)$	0.30	0	0.3	1	0.06
	On-demand Flow Rate	Qin	$S(Q_D) = \frac{1}{2} \left(1 - \cos \left(\frac{Q_D - Q_{D\min}}{Q_{D\max} - Q_{D\min}} \pi \right) \right)$	29.96	18	40	0.5686559	0.035533369
	Reliability (days consumption met),	R	$S(R) = \frac{1}{2} \left(1 - \cos \left(\frac{R - R_{\min}}{R_{\max} - R_{\min}} \pi \right) \right)$	365.00	200	365	1	0.14
	Overall Satisfaction						87%	88%

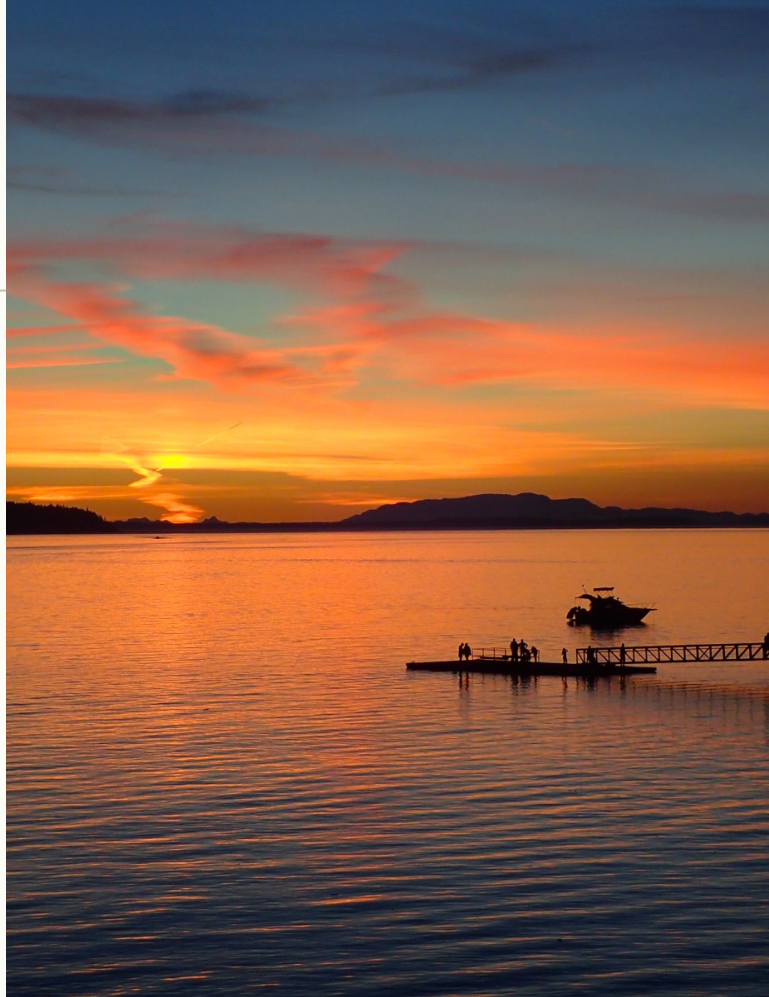
03

Final Recommendation

Conclusion

Our solution provides:

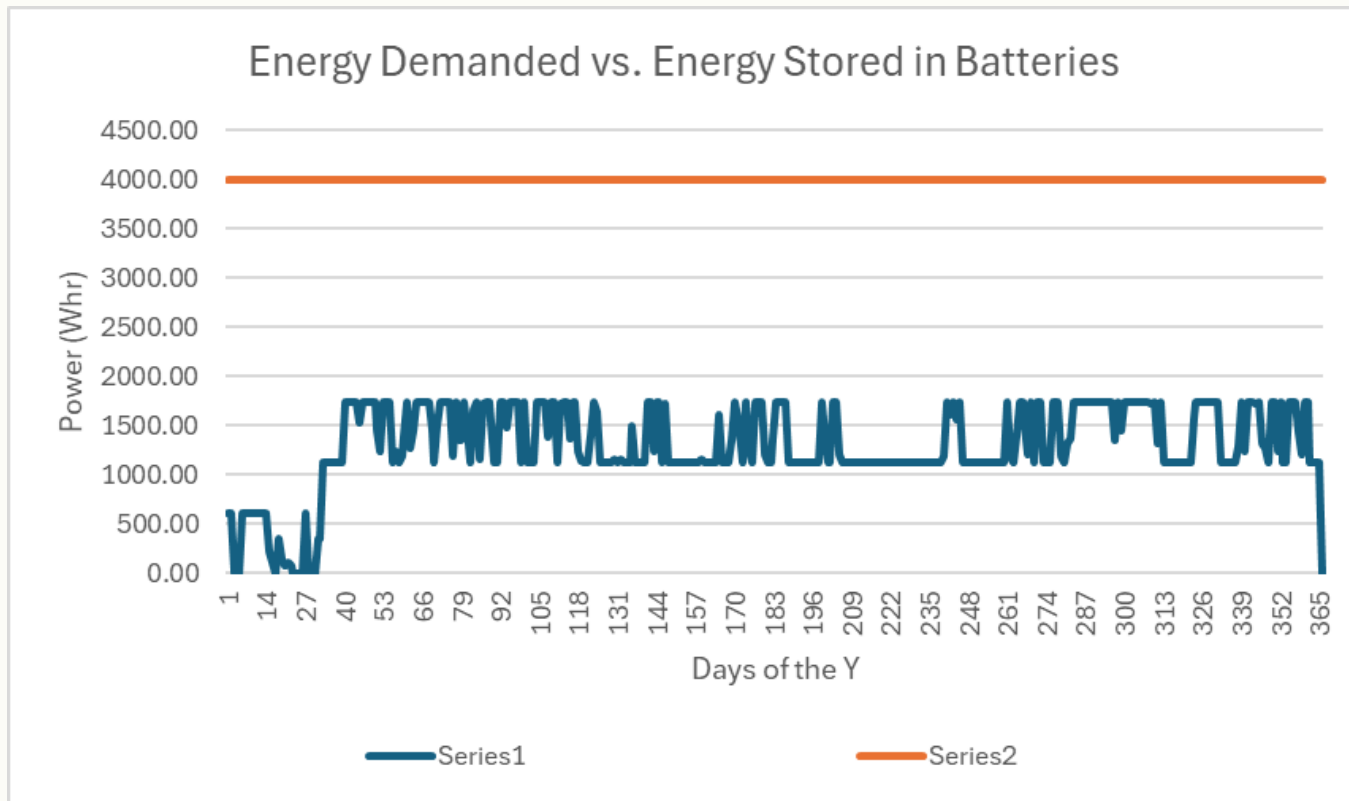
- Plentiful Consumption
- Mid Range cost
- Low health and environmental risk
- 365 day reliability



Thanks!

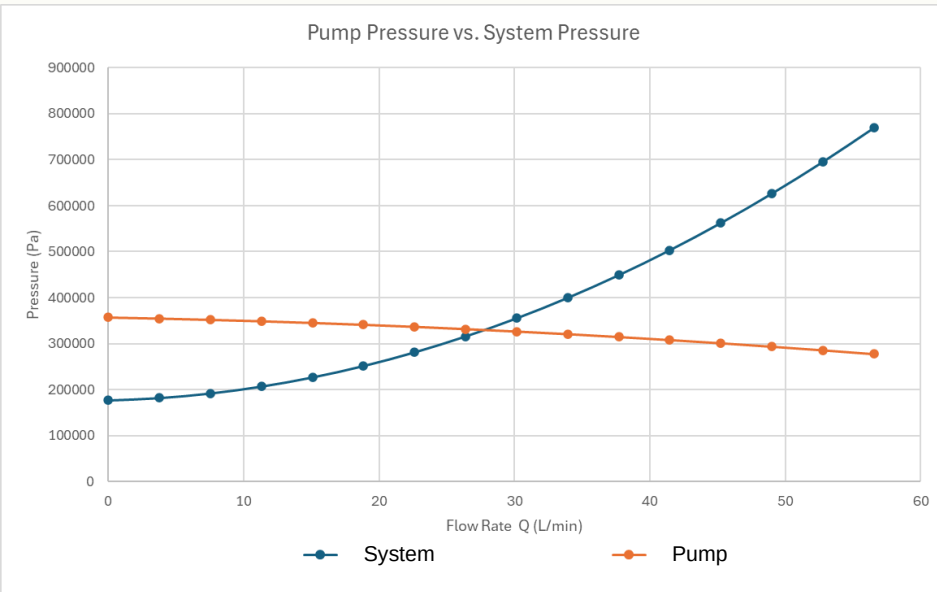
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Appendix A- Energy in the solar panel and batteries

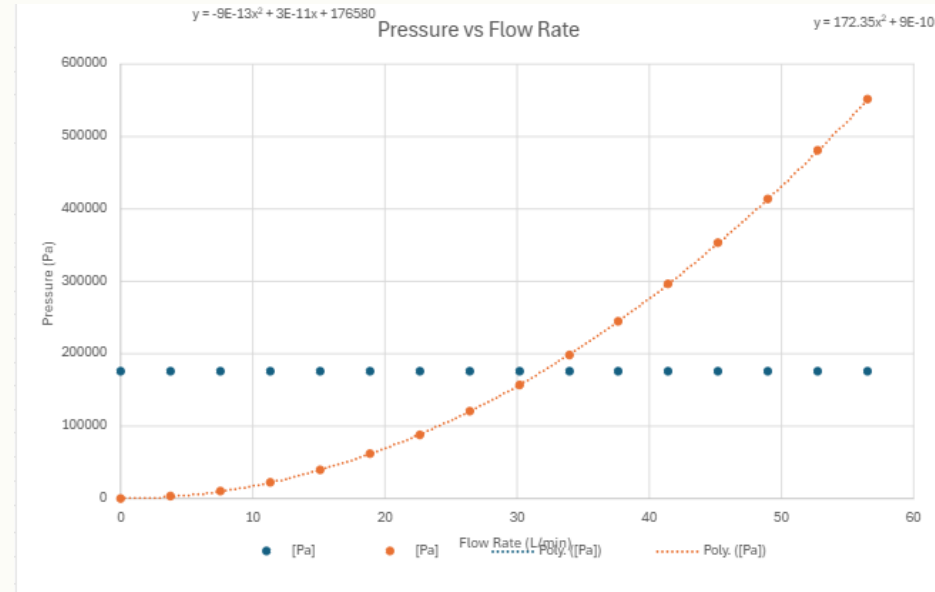


Appendix B- System curve and pump curve

House to Storage



Storage to House



Credits

1. Images of components were taken directly from provided component selection guide:

https://canvas.ubc.ca/courses/176498/files/45589074?module_item_id=9106435

Contributions

Ruoyu	Slides 1-4
Clint	Slides 5-9
Claus	Slides 10-12
Michael	Slides 13-17
Sameen	Slides 17-22
Jack	Slides 23-29

Use of GenAI

OpenAI. (2025). *ChatGPT* (GPT-5) [Large language model]. <https://chatgpt.com/>

Our uses of GenAI:

- Improve grammar and check spelling
 - Improve slide formatting
- Improve image clarity/sharpness
 - Generate starred images (*)